Postulates & Diagrams — Practice

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Which Postulate?

Watch Me — Worked Example

"Points P and Q lie in plane M, so line PQ lies in plane M."
Postulate: if two points lie in a plane, then the **line containing them** lies in that plane.
Other postulates: two points $\rightarrow$ one line; three noncollinear points $\rightarrow$ one plane; two lines meet in one point; two planes meet in one line.

1. How many lines pass through points A and B?

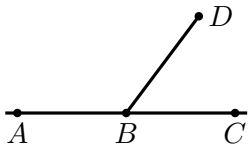
2. How many planes contain three noncollinear points J, K, L?

3. Two different planes intersect. What is their intersection?

4. Two different lines intersect. What is their intersection?

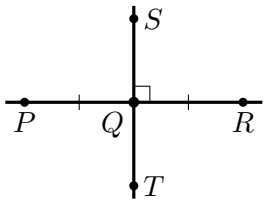
Type 2 What Can You Assume From a Diagram?

Watch Me — Worked Example



Yes: A, B, C are collinear; $\angle ABD$ and $\angle DBC$ are a linear pair (drawn).
No: $\angle DBC = 60°$, or $AB = BC$ — no marks say so.

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Use the diagram. Write **yes** or **no**.

5. P, Q, R are collinear. _____

6. $\overline{PR} \perp \overline{ST}$. _____

7. $PQ = QR$. _____

8. $SQ = QT$. _____

Type 3 Sketch It

Watch Me — Worked Example

"Sketch two planes that intersect in line ℓ." Draw two flat sheets crossing each other and label the crossing line ℓ.
Tip: use tick marks and right-angle marks only for things that are given.

9. Sketch line m intersecting plane X at point A.

10. Sketch $\overline{AB}$ with midpoint M (use tick marks).

Type 4 Always, Sometimes or Never?

Watch Me — Worked Example

"Three points are coplanar." Through any three points there is at least one plane. **Answer: always.**
"Three points are collinear." Some are on one line, some are not. **Answer: sometimes.**

11. Two points are collinear.

12. Two lines intersect in two points.

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13. Four points are coplanar.

14. Two planes intersect in a point only.

15. A line and a plane intersect in one point.

16. HARD Through a line and a point not on it, how many planes pass?

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